

DENTON, TX, USA

WMO: 722589

Lat: 33.206N

Lon: 97.199W

Elev: 196

StdP: 99.00

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 03991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-4.1	-12.9	1.3	-2.2	-10.7	1.5	1.1	12.0	12.0	11.1	11.7	3.4	350	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	38.7	23.5	37.4	23.9	36.2	23.9	25.9	33.8	25.5	33.3	25.0	32.7	4.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.3	28.9	23.3	18.6	28.3	22.9	18.0	27.9	81.3	33.8	79.0	33.6	76.9	32.6	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.5	8.7	-10.4	40.2	2.4	1.8	-12.2	41.6	-13.6	42.6	-14.9	43.7	-16.7	45.0
			-11.2	26.8	2.2	0.7	-12.7	27.3	-14.0	27.7	-15.2	28.0	-16.8	28.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	7.0	9.3	13.8	17.9	22.7	27.3	29.4	29.5	25.3	19.0	13.1	7.8
	DBStd	9.14	5.47	6.05	5.36	4.36	3.80	2.49	2.30	2.57	3.53	4.82	5.31	5.40
	HDD10.0	370	124	80	24	2	0	0	0	0	2	29	108	
	HDD18.3	1384	352	258	157	59	10	0	0	0	2	51	168	328
	CDD10.0	3496	32	59	143	241	395	518	601	604	458	282	123	40
	CDD18.3	1469	1	4	17	47	146	268	342	346	210	72	12	2
	CDH23.3	17547	12	34	133	398	1418	3148	4583	4644	2346	720	97	13
Wind	WSAvg	8940	1	4	22	89	528	1578	2625	2694	1136	244	17	2
	WSAvg	4.1	4.1	4.5	4.8	4.9	4.6	4.2	3.7	3.4	3.3	3.8	4.0	4.0
Precipitation	PrecAvg	916	57	61	83	99	119	89	56	51	66	95	77	64
	PrecMax	1409	144	188	178	229	324	298	235	127	251	268	251	226
	PrecMin	423	3	5	0	13	13	12	0	0	1	0	6	7
	PrecStd	248	36	47	48	49	65	62	51	37	53	74	63	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	27.1	29.0	31.6	35.2	37.5	40.2	41.1	38.2	33.6	28.7	25.1
		MCWB	14.6	16.4	18.3	21.4	23.9	24.3	23.8	23.6	22.8	21.9	18.4	16.9
	2%	DB	22.2	23.8	26.7	28.8	32.9	36.1	38.6	39.0	35.9	31.6	26.1	22.2
		MCWB	14.5	16.1	18.0	20.1	23.0	24.1	23.7	23.4	22.8	21.5	17.8	15.1
	5%	DB	19.2	21.8	24.6	27.4	31.3	34.7	37.4	37.7	34.0	29.2	23.8	19.8
		MCWB	13.2	15.2	17.1	19.7	22.6	23.8	24.1	23.8	22.7	20.2	17.2	14.8
	10%	DB	17.1	19.2	22.5	25.8	29.6	33.5	36.0	36.4	32.5	27.5	22.0	17.3
		MCWB	11.9	14.2	16.2	18.8	21.7	23.7	24.0	23.9	22.5	19.6	16.4	12.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	19.6	20.8	23.5	25.8	27.1	26.7	26.2	25.6	24.3	21.3	19.4
		MCDB	21.0	23.4	26.1	27.9	32.8	34.8	35.0	34.4	32.7	29.2	25.7	22.4
	2%	WB	16.6	18.4	19.7	22.1	24.6	25.8	25.9	25.6	24.8	23.2	19.7	17.7
		MCDB	19.6	21.7	24.3	26.7	30.3	33.1	34.0	34.1	31.7	28.7	23.2	20.9
	5%	WB	14.6	16.9	18.6	21.1	23.9	25.1	25.5	25.1	24.1	22.3	18.7	15.6
		MCDB	18.2	20.0	22.8	25.3	29.0	32.0	33.5	33.8	30.6	27.3	22.2	18.1
	10%	WB	12.0	15.1	17.6	20.1	23.0	24.6	25.0	24.7	23.6	21.2	17.3	13.2
		MCDB	16.3	18.9	21.4	24.1	28.2	31.1	32.8	32.9	29.8	25.6	21.1	16.8
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.2	12.1	12.2	11.4	11.1	11.9	12.1	12.2	12.8	12.4	11.9
		MCDBR	16.9	15.2	14.0	13.5	12.7	12.5	13.5	13.8	13.5	14.3	14.1	14.9
	5% WB	MCWBR	10.5	9.3	7.5	7.2	5.3	4.0	3.7	3.7	4.4	6.3	7.9	9.2
		MCDBR	14.1	12.9	11.4	10.7	10.7	10.8	11.7	11.9	11.0	11.1	11.7	11.4
		MCWBR	10.3	9.3	7.2	6.9	5.4	4.2	3.9	3.7	4.1	5.5	8.4	9.0
Clear-Sky Solar Irradiance	taub		0.312	0.324	0.356	0.389	0.425	0.437	0.438	0.438	0.397	0.358	0.333	0.317
	taud		2.496	2.472	2.400	2.333	2.296	2.296	2.329	2.313	2.374	2.470	2.479	2.482
	Ebn at Noon		907	931	921	900	866	851	848	843	866	880	873	878
	Edh at Noon		86	97	113	126	133	132	127	127	115	97	87	83
All-Sky Solar Radiation	RadAvg		2.88	3.41	4.48	5.44	5.87	6.60	6.71	6.16	5.15	4.03	3.06	2.52
	RadStd		0.29	0.53	0.33	0.44	0.58	0.64	0.45	0.38	0.47	0.60	0.39	0.31

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page